

From suffixes to particles: Comparative constructions in Tsugni Dargwa and other East Caucasian languages

Abstract:

This paper presents a microtypological study of comparative constructions in the East Caucasian languages, with special attention to Tsugni Dargwa, a one-village lect within the Dargwa (Dargwic) group. The main research question of this study concerns the type(s) of comparative constructions used in each language, in terms of Leon Stassen's classification. We found that although most East Caucasian languages have Adverbial (Locational) comparative constructions, many of them have Particle comparatives. In most cases, Particle comparatives coexist with Locational constructions, and in many languages, they have developed from earlier Locational comparatives. Morphological aspects of this complex process can be viewed in terms of degrammaticalization.

Keywords:

comparative constructions, standard of comparison, East Caucasian languages, Dargwa, Tsugni

1 Preliminary remarks

This paper describes comparative constructions in the East Caucasian (Nakh-Dagestanian) languages. To be more precise, it concentrates on comparative construction types, in terms of Stassen (1985; 2001; 2013; see Section 2) and on the development of so-called Particle comparatives in the languages where these constructions are found.

Our research is a pilot study, which is mainly based on published works; where possible we also consulted colleagues and, in a few cases, native speakers of several East Caucasian languages. One particular East Caucasian lect, Tsugni Dargwa, is described in more detail. The Tsugni data were obtained via elicitation and introspection (one of the authors, [...], is a native speaker of Tsugni). Additionally, we used materials from a corpus of oral texts in the Tsugni lect compiled by [...], totaling approximately 7,000 words.

We express our deepest gratitude to our colleagues and native speakers of several East Caucasian languages who shared their knowledge with us: [acknowledgements to be inserted later].

The paper is structured as follows. Section 2 introduces the typological classification of comparative constructions on which our research is based, along with the terminology that is used to describe these constructions. Section 3 provides information on the East Caucasian languages that is essential for understanding the subsequent analysis. Section 4 is a case study: it describes comparative constructions in Tsugni Dargwa and demonstrates the basic features distinguishing Adverbial vs. Particle comparatives. In Section 5, we present a survey of comparative constructions across the East Caucasian language family. Section 6 contains the results of the study.

2 Stassen's classification of comparative constructions

As noted by Leon Stassen (2013), "comparison can be defined as a mental act by which two objects are assigned a position on a predicative scale". The positions of the two objects can be different (*John is smarter than Sam*) or the same (*John is as smart as Sam*). When such relations are employed by special syntactic constructions, we can refer to these constructions as *comparative* and *equative*

constructions, respectively¹. Comparative constructions typically include the following elements (Cuzzolin & Lehmann 2004: 1212):

- “(a) the element which is compared, the **comparee**, or **topic of comparison**;
- (b) the element that serves as **standard of comparison**;
- (c) the element, called the **pivot**, or **marker of comparison**, that introduces the standard of comparison;
- (d) the predicate that represents the parameter of comparison.”

In this paper, we are mainly interested in markers of comparison, or *pivots*, in terms of (Cuzzolin & Lehmann 2004). Other terms referring to the same phenomenon are *mark of the grammatical function of the Standard* (Dixon 2008), *tie* (Stolz 2013: 9), *Standard Marker* (Treis 2018). We will use the term *standard marker*. For instance, in example (1) from Cuzzolin & Lehmann (2004), *John* is the comparee, *Sam* is the standard of comparison, *smarter* is the parameter of comparison, and *than* is the standard marker:

(1) *John is smarter than Sam.*

A well-known typological classification of comparative constructions, introduced in the works of Leon Stassen (1985; 2001; 2013), is based on the expression of the standard of comparison. This classification divides comparative constructions into two main types, each with two subtypes.

In *Fixed-case comparatives*, the standard of comparison invariably takes the same grammatical form. These constructions are further divided into two subtypes: those where the standard of comparison is introduced by an additional predicate with the meaning ‘exceed’ or ‘surpass’ (*Exceed comparatives*) and those in which the standard of comparison directly depends on the main predicate of the sentence and is marked by a particular case form or adposition. The latter type, which is especially important for us, Stassen (2013) calls *Locational comparatives*, because in this type the standard of comparison most usually takes the form of a locative case. However, some languages have comparative constructions of the same type, but the standard marker is a dedicated comparative suffix. That is why we will refer to this class as *Adverbial comparatives* (the term from Stassen 1985²).

Example (2) from Landuma (Mel < Niger-Congo) illustrates an Exceed comparative, and sentence (3) from Lezgian an Adverbial comparative.

Landuma (Mel < Niger-Congo)

(2) *ŋä-n-tana* *lɛ* *ŋä-tas* *mi*,
 3PL.AN-PFV-be.small ASR 3PL.AN-exceed 1SG.O
 min *m-a* *ŋä* *mi* *fəna* *ma-tana*.
 heart CL-POSS 3PL.AN.O CL-DEF also CL-be.small
 [The cat thought:] ‘They [mice] are smaller than me, and their minds [lit. ‘heart’] are small too.’

Lezgian

(3) *dedi-di* *tort-ar* *xala-di-laj* *ijer-diz* *čra-da*.
 mother-OBL(ERG) cake-PL aunt-OBL-SUPER.EL good-ADV bake-FUT
 ‘Mother bakes cakes better than aunt.’ (Haspelmath 1993: 433)

¹ Another possible solution is to use the term *comparative constructions* both for constructions expressing different positions of the two objects on the scale (comparison of inequality) and constructions expressing equal positions on the scale (comparison of equality); cf., for example, Stolz (2013).

² “[I]n the typical case of this category, the comparative predicate is the only predicative form in the construction, and that is modified syntactically by the adverbial phrase containing the standard NP” (Stassen 1985: 31).

In *derived-case comparative constructions*, the form of the standard of comparison depends on the form of the comparee. These constructions may involve coordination, where two clauses are contrasted (such as *X is big, Y is small* meaning ‘X is bigger than Y’; example 4), or constructions where a specific marker (a particle or conjunction) introduces the standard of comparison, and the standard of comparison shares the same case as the comparee. Constructions with comparative particles are typical of European languages (5).

(4) Samoan
Ua loa lenet va’a, ua puupuu lena.
 is long this boat is short that
 ‘This boat is longer than that boat.’ (Stassen 1985: 38)

(5) Russian
Masha ljubit sobak bol’she, chem koshek.
 PN love:PRS.3SG dog:PL.ACC more than cat:PL.ACC
 ‘Masha loves dogs more than cats.’

Below, when discussing comparative constructions in the East Caucasian languages, we use certain terminological simplifications. In a comparative construction, the comparee and the standard of comparison are syntactic constituents, not just lexemes. However, we usually speak about them as words, meaning the lexical head of the corresponding constituent. We say, for example, that the standard of comparison is a noun, an adverb or a verb form instead of saying that it is a noun phrase, an adverb phrase, etc. Following the terminology in Stassen (1985; 2013), we also use the term *particle* when referring to the standard marker in Particle comparatives, although syntactically this element should rather be viewed as a conjunction.

3 East Caucasian languages: necessary information

3.1 General

The East Caucasian (Nakh-Dagestanian) language family comprises more than 30 languages spoken in the eastern parts of the Caucasus (mainly in the Russian Federation, but also in Azerbaijan and Georgia). A classification of the East Caucasian languages (Alekseev 2001) is given in (6); the Dargwa (Dargwic) languages are shown according to Koryakov (2021).

(6) Classification of East Caucasian languages (Alekseev 2001)

NAKH	Batsbi, Chechen, Ingush
AVARO-ANDO-TSEZIC	
AVAR	Avar
ANDIC	Akhvakh, Andi, Bagvalal, Botlikh, Chamalal, Godoberi, Karata, Tindi
TSEZIC	Bezhta, Hunzib, Hinuq, Khwarshi, Tsez
LAK	Lak
DARGWA	Northern Dargwa, Tsudaqar, Usisha-Butri, Gapshima, Muira, Mehweb, South-Western Dargwa (including Tsugni, Tanti, etc.), Sanzhi-Itsari, Sanakari-Chakhrizhi, Amuzgi-Shiri, Kubachi, Ashti, Chirag, Kaytag, Shari
LEZGIC	Agul, Archi, Budukh, Kryz, Lezgian, Rutul, Tabasaran, Tsakhur, Udi
KHINALUG	Khinalug

The East Caucasian languages are morphologically ergative, although many syntactic processes follow the nominative-accusative pattern. They are essentially left-branching, with rather free word order. They have rich inflectional morphology in both the nominal and verbal domains. Most East Caucasian

languages have well-developed gender agreement. Some languages (Lak, Dargwa, Udi, Tabasaran, etc.) also show agreement in person. Dependent clauses are normally headed by non-finite verb forms: participles in relative clauses; various converbs in adverbial clauses; infinitives, deverbal nouns or converbs in complement clauses. Finite verb forms in dependent clauses are allowed only in sentences conveying other people's speech.

3.2 Declension

Most East Caucasian languages have a two-stem declension system. As explained in Daniel & Ganenkov (2009: 670): "Declension mostly follows a two-stem pattern: all cases except nominative³ are derived from a common stem called oblique, while the nominative is derived from a direct stem and is most often formally identical to it (thus being zero-marked). The oblique stem is derived from the direct stem by adding various morphemes called oblique stem markers".

Example (7) from Lezgian shows some forms of the noun *balk'an* 'horse'; the oblique stem marker is *-di* in the singular and *-i* in the plural.

Lezgian (Daniel, Ganenkov 2009: 671)

(7)	a.	<i>balk'an</i> horse	b.	<i>balk'an-di-</i> horse-OBL-	c.	<i>balk'an-di-z</i> horse-OBL-DAT
	d.	<i>balk'an-ar</i> horse-PL	e.	<i>balk'an-ar-i-</i> horse-PL-OBL-	f.	<i>balk'an-ar-i-z</i> horse-PL-OBL-DAT

In Tsugni Dargwa, as can be seen from (8), the oblique stem marker in the singular is *-li*; this marker is identical to the ergative case suffix. In the plural, the oblique marker is *-a*.

Tsugni

(8)	a.	<i>k:alk:a</i> tree	b.	<i>k:alk:a-li</i> tree-ERG	c.	<i>k:alk:a-li-j</i> tree-OBL-DAT	d.	<i>k:alk:a-la</i> tree-GEN
	e.	<i>k:alk-me</i> tree-PL	f.	<i>k:alk-m-a-li</i> tree-PL-OBL-ERG	g.	<i>k:alk-m-a-j</i> tree-PL-OBL-DAT	h.	<i>k:alk-m-a-la</i> tree-PL-OBL-GEN

It is common to find that some nouns in a given language derive all of their case forms from the direct stem. Besides, in some languages there are certain case forms which are derived from the direct stem for all nouns, such as the genitive singular in Tsugni (8d).

The nominal case forms can be divided into two groups: non-locative (grammatical) forms and locative (spatial) forms. The most common grammatical cases are absolutive (usually unmarked), ergative, dative and genitive; many languages also have comitative, instrumental, affective, etc. (Daniel & Ganenkov 2009).

The locative forms generally express two or, less frequently, three different categories at once — localization, orientation (movement), and direction — and have two or three corresponding morphological slots for the suffixes expressing these categories. Localization specifies a spatial area (= reference point) in relation to the object denoted by the noun (inside the object, in front of it, behind it, etc.). Orientation indicates whether the object in question is localized at the reference point (essive) or moves with respect to it (lative, movement to the reference point; elative, movement away from the reference point⁴; allative, movement in the direction of the reference point without reaching it; translative, movement across the reference point, etc.). Lastly, the category of direction orients movement with respect to the speaker (toward or away from the speaker) or on the vertical axis (up or down). It cannot be defined for essive forms.

³ We use the term *absolutive*.

⁴ In East Caucasian linguistics, different authors use the terms *ablative* or *elative* for the same orientation; in this paper, we consistently use the term *elative*.

Example (9) presents some locative forms from Agul. This language has two locative categories, localization (in our example ‘behind’ and ‘in front’) and orientation (essive – lative – elative).

Agul: examples of locative forms of the noun *χal* ‘house’ (Daniel, Ganenkov 2009: 675)

- (9) a. *χul-a-q*
house-OBL-POST(ESS)⁵
‘behind the house’
d. *χul-a-h*
house-OBL-ANTE(ESS)
‘in front of the house’
b. *χul-a-q-t:i*
house-OBL-POST-LAT
‘(to) behind the house’
e. *χul-a-h-t:i*
house-OBL-ANTE-LAT
‘(to) in front of the house’
c. *χul-a-q-as*
house-OBL-POST-EL
‘from behind the house’
f. *χul-a-h-as*
house-OBL-ANTE-EL
‘from in front of the house’

Table 1 contains a list of locative categories and their markers in Tsugni Dargwa; this lect has three locative categories. Examples of locative forms in Tsugni are in (10).

Table 1. Markers of locative forms in Tsugni

Localization		Orientation		Direction	
SUPER ‘on’	-ge	lative	unmarked	‘downward’	-ka
IN ‘in (a container)’	-le	essive	-GM ⁶	‘upward’	-ha
INTER ‘in (a substance), among’	-c:e	elative	-r	‘to the speaker’	-ca
SUB ‘under’	-gu	allative	-GM-a	‘from the speaker’	-de
AD ‘at’	-š:u				
ANTE ‘in front of’	-sa				
POST ‘behind’	-gira				

Tsugni Dargwa: examples of locative forms of the noun *q’wani* ‘chest’

- (10) a. *q’wani-li-ge*
chest-OBL-SUPER(LAT)
‘onto the chest’
c. *q’wani-li-ge-r-ha*
chest-OBL-SUPER-EL-UP
‘up from the chest’
e. *q’wani-li-ge-b-a-ka*
chest-OBL-SUPER-N-ALL-DOWN
‘down in the direction onto the chest’
g. *q’wani-le-b*
chest-IN-N(ESS)
‘in the chest’
b. *q’wani-li-ge-b*
chest-OBL-SUPER-N(ESS)
‘on the chest’
d. *q’wani-li-ge-r-ca*
chest-OBL-SUPER-EL-HITHER
‘hither from the chest’
f. *q’wani-le*
chest-IN(LAT)
‘into the chest’
h. *q’wani-le-r-ha*
school-IN-EL-UP
‘up out of the chest’

The same or similar declension patterns are usually characteristic of pronouns, adjectives and participles (when used without a nominal head).

In addition, partial declension is characteristic of locative adverbs, which inflect for orientation and, in some languages, direction. Example (11) contains several forms (essive – elative – translative) of the locative adverbs ‘below’ and ‘above’ in Bagvalal, while (12) gives lative, essive and elative forms of the adverb ‘outside’ in Tsugni.

⁵ The essive orientation is unmarked.

⁶ In Tsugni, the essive orientation is marked by the presence of a gender marker after the localization marker. Gender is cross-referenced on many targets belonging to different lexical classes, including the essive and allative forms of nouns. In the most common case, gender agreement is controlled by the absolutive argument of the same clause; cf. examples in Section 4.2.

- 1
2 Bagvalal (Sosenskaja 2001b: 170)
- 3 (11) a. *hiL'i* b. *hiL'i-s:* c. *hiL'i-s:ini*
4 under(ESS) under-EL under-TRANS
5 'under, below' 'from under' 'through the space under (something)'
- 6 d. *č'ihi* e. *č'ihi-s:* f. *č'ihi-s:ini*
7 above(ESS) above-EL above-TRANSL
8 'on, above' 'from (above)' 'through the space over (something)'
- 9
- 10 Tsugni Dargwa
- 11 (12) a. *t:ura* b. *t:ura-b* c. *t:ura-r-ka* d. *t:ura-b-a-de*
12 outside outside-N outside-EL-DOWN outside-N-ALL-THITHER
13 'outward' 'outside'(position) 'from outside down' 'in the direction of outside'

14 3.3 Comparative constructions

15 The most common type of comparative constructions found in East Caucasian languages are Adverbial
16 comparatives. In a typical Adverbial comparative, the adjective or adverb expressing the parameter of
17 comparison has no special marking, while the noun phrase representing the standard of comparison
18 appears in a locative case. The standard of comparison is most frequently (but not exclusively) marked
19 by one of the elative forms (for more information, see Naccarato 2020). The case of the standard does
20 not depend on the form of the comparee.

21 Examples (13) and (14) illustrate the standard of comparison in the super-elative case in
22 Bagvalal and in the elative form of the default localization in Sanzhi Dargwa.

- 23
- 24 Bagvalal
- 25 (13) *jaš-t:i-r* *waša-š:a-s:* *beʃila* *bihi-r-o:-b* *keč'.*
26 girl-OBL-ERG boy-OBL.M.SUPER-EL well take-IPFV-PTCP-N song(ABS)
27 'The girl is singing the song better than the boy' (super-elative). (Sosenskaja 2001a: 409–410)
- 28
- 29 Sanzhi Dargwa
- 30 (14) *u-le-rka* *sala-r* *du-l* *maχ* *χ:ula-ce* *b-arq'-ij.*
31 you.SG-LOC-EL front-EL I-ERG barrow(ABS) big-ADJ.SG N-do.PFV-INF
32 'I (will) make a big barrow earlier than you' (general localization, elative). (Forker 2020: 569)

33

34 The comparative constructions illustrated in (3) and (13)–(14) share the following features:

35 (i) the adjective or adverb expressing the parameter of comparison does not have any special
36 comparative forms (no comparative/superlative degree),

37 (ii) the NPs encoding the comparee and the standard of comparison are in the same clause,

38 (iii) the standard of comparison is expressed by a locative case,

39 (iv) the locative case markers are suffixes that usually attach to oblique stems.

40

41 There is no detailed typological study of comparative constructions in the East Caucasian family or
42 across the whole Caucasian region. We only have data on separate languages, which are in many cases
43 very scarce, and the database Naccarato (2020), which contains information on standard markers.
44 According to this database, there are two types of standard markers in East Caucasian: *spatial* (locative)
45 and *dedicated* markers. The languages with spatial markers are much more numerous (29 out of 38
46 East Caucasian lects in the database); by default, we expect the corresponding comparative
47 constructions to belong to the Adverbial type (Section 3), but this is not the only option. Nine
48 languages have dedicated standard markers: Archi, Budukh, Rutul, Tabasaran, Hunzib, Khinalug, Lak,
49 Chechen and Ingush.

In our study of East Caucasian comparative constructions, the research question was different from that addressed in Naccarato (2020). We aimed to identify the type of comparative constructions found in each language, in terms of Stassen's classification. As it turns out, the choice is between Adverbial and Particle comparatives only; other types of comparative constructions are not attested in the family.

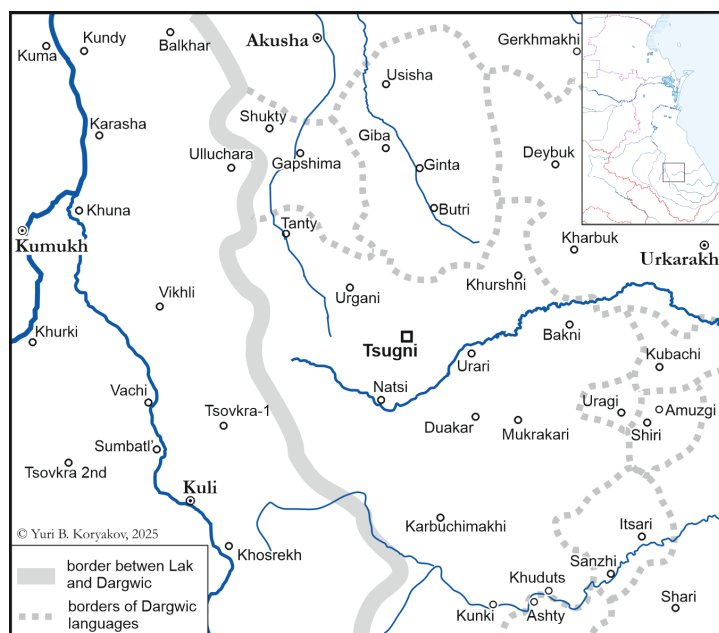
To distinguish between the two types of comparative constructions, we had to identify whether these constructions were monoclausal or biclausal and whether the standard marker was a case suffix (or a chain of 2–3 suffixes) or a particle. This is not always trivial, because in the East Caucasian languages the standard marker is in most cases a phonologically bound element that immediately follows the standard of comparison (examples 3, 13, 14). Moreover, in most descriptions the syntactic structure of comparatives is not discussed. That is why we singled out several important features of comparative constructions that, on the one hand, help to differentiate between Adverbial and Particle comparatives and, on the other, can usually be extracted from descriptive works.

To demonstrate these features, we will start with a case study – a description of comparative constructions in Tsugni Dargwa. This is a lect that makes use of Particle comparative constructions, as we attempt to show below. We will emphasize the differences between Particle and Adverbial comparative constructions by comparing Tsugni to Tanti Dargwa, a closely related lect with Adverbial comparatives.

4 A case study: Comparative constructions in Tsugni Dargwa

4.1 Tsugni Dargwa: General information

Tsugni Dargwa is a one-village lect spoken in the village of Tsugni, which is located in the Akushinsky District of the Republic of Dagestan (Russian Federation). Currently, the village has a population of around 300 inhabitants; however, the total number of Tsugni speakers is considerably higher (estimated at 1,500 to 5,000), since the language is still maintained by those who have migrated to urban areas and lowland rural settlements in Dagestan and other regions of Russia.



Map 1. Tsugni and neighboring villages

Tsugni belongs to Dargwa (Dargwic), a distinct branch of the East Caucasian family. According to Yuri Koryakov's classification (2021), Tsugni is a lect of South-Western Dargwa; see classification (6) in

Section 3.1. We will compare the data of Tsugni to Tanti, another one-village lect belonging to South-Western Dargwa, spoken in the same area (Sumbatova & Lander 2014; Lander & Mamaev 2021).

The standard of comparison in Tsugni takes markers that are identical to elative markers *-le-r-ka* IN-EL-DOWN and *-le-r-ha* IN-EL-UP⁷, cf. (15). In Section 4.2, until we clarify the morphological status of these markers in the comparative function (suffixes vs. clitics), we write them with dots (*le.r.ka* and *le.r.ha*) and gloss them as COMP in square brackets:

Tsugni Dargwa

- (15) *hinc-be quʳr-be-le.r.ka / -le.r.ha d-urqa-ce ca<d>i.*
 apple-PL(ABS) pear-PL[-COMP] / [-COMP] NPL-expensive-ADJ COP<NPL>
 ‘Apples are more expensive than pears.’

Examples (16) and (17) illustrate these markers in their original spatial meaning:

Tsugni Dargwa

- (16) *t:at:i uškul-le-r-ka quli če-sa-w-ačʼ-ib.*
 father(ABS) school-IN-EL-DOWN home ON-HITHER-M-COME.PFV-AOR
 ‘Father returned [down] home from school’ [the school is situated higher than the father’s house].

- (17) *s:ika-la gal-li nemcʼ-t:ura-b-arqʼ-ib-le ca-j liil waza qaba-le-r-ha.*
 bear-GEN child-ERG lick-outside-N-do.PFV-AOR-CVB COP-M all<N> honey(ABS) pitcher-IN-EL-UP
 ‘The bear cub licked [up] all the honey out of the jug.’

Although the initial spatial meanings of *le.r.ka* and *le.r.ha* are different, in the comparative function they are in free variation.

4.2 Comparative constructions

4.2.1 Basic comparative constructions

In many descriptive works, comparative constructions are illustrated by sentences where the comparee is an absolutive NP, like (15). When discussing such constructions, we will use the term ABS(olutive)-comparison.

A single example of ABS-comparison is not enough to establish the type of comparative construction used in a given variety. If we look at sentences (18ab) from Tsugni and Tanti, for example, we observe seemingly parallel structures.

- (18) Tanti Dargwa (Sumbatova & Lander 2014: 73)

- a. *hit xuri-ja-r haq-le sa-j.*
 that(ABS) rock-SUPER-EL high-ADV COP-M

Tsugni Dargwa

- b. *it qir-le.r.ka aq-le ca-j.*
 that(ABS) rock[-COMP] high-ADV COP-M
 (a=b) ‘He is taller than the rock.’

⁷ On occasion Tsugni speakers may use different standard markers: *=leher*, *=leken*, *=kalen*. Synchronically, *=leher*, *=leken* and *=kalen* do not coincide with any locative markers, although they partly consist of the same elements (*-le*, *-r*, *-ka*, *-ha*). However, the speakers explicitly prefer the markers *=lerka* and *=lerha* and tend not to accept examples with *=leken* and *=kalen*. In what follows, we will discuss only the markers *=lerka* and *=lerha*.

In both lects, the standard of comparison takes the form of a locative case (or at least coincides with a locative case form): in Tanti this is the super-relative, in Tsugni the in-relative.

Examples of ABS-comparison can nonetheless be useful for identifying the construction type, but we will need more examples with different comparees.

First, in Tsugni, the localization IN (with the suffix *-le*) has certain limitations: it cannot be formed from animate nouns and from personal, demonstrative and other pronouns. This restriction also applies to elative forms marked by *-le-r-ka* and *-le-r-ha*, cf. examples of grammatical and ungrammatical forms in (19)–(21).

Tsugni Dargwa								
(19)	a.	<i>q'wani-le</i>	b.	<i>q'wani-le-b</i>	c.	<i>q'wani-le-r-ka</i>	d.	<i>q'wani-le-r-ha</i>
		chest-LOC(LAT)		chest-LOC-N(ESS)		chest-LOC-EL-DOWN		chest-LOC-EL-UP
		'into the chest'		'inside the chest'		'down out of the chest'		'up out of the chest'
(20) ⁸	a.	<i>*admi-le</i>	b.	<i>*admi-le-b</i>	c.	<i>*admi-le-r-ka</i>	d.	<i>*admi-le-r-ha</i>
		person-LOC(LAT)		person-LOC-N(ESS)		person-LOC-EL-DOWN		person-LOC-EL-UP
		'into the person'		'inside the person'		'down out of the person'		'up out of the person'
(21)	a.	<i>*il-le</i>	b.	<i>*il-le-b</i>	c.	<i>*il-le-r-ka</i>	d.	<i>*il-le-r-ha</i>
		this-LOC(LAT)		this-LOC-N(ESS)		this-LOC-EL-DOWN		this-LOC-EL-UP
		'into this'		'inside this'		'down out of this'		'up out of this'

By contrast, the standard markers *le.r.ka* and *le.r.ha* can attach to any noun, as well as to personal and demonstrative pronouns, as in *admi-le.r.ka* 'than a person', *du-le.r.ka* 'than me' or *ij-le.r.ka* 'than this'; cf. (24), (26), (29a) below.

Our second consideration concerns the choice of the nominal stem. The localization IN is derived from the direct stem in the singular, but from the oblique stem in the plural, cf. *ʔačn-a-le-r-ka* 'from works' in (22). But the standard markers *le.r.ka* and *le.r.ha* consistently attach to the direct plural stem (or, rather, to the plural absolutive form), cf. *qʷe-d-ce ʔačn-e-le.r.ka* 'than works in the field' (23).

Tsugni Dargwa				
(22)	<i>ʔač-n-a-le-r-ka</i>	<i>če-sa-ʔaʷ-b-ak'-a-j</i>	<i>nalla.</i>	
	work-PL-OBL-LOC-EL-DOWN	ON-ANTE-NEG-HPL-COME.PFV-AOR.NEG-3	still	
	'[They] haven't come back from work yet' (lit. 'from works')			
(23)	<i>qʷe-d-ce</i>	<i>ʔač-n-e-le.r.ka</i>	<i>q:ijan-ce</i>	<i>ce-k'al</i>
	field-NPL-ADJ	work-PL[-COMP]	hard-ADJ	what-INDF(ABS)
	N-NEG.COP(3)			
	'There's nothing harder than fieldwork.' (lit. 'works in the field')			

This means that in the comparative function, *le.r.ka* and *le.r.ha* do not follow the selectional restrictions typical of locative case markers. They rather attach to whole nominal forms (in the absolutive case).

If we look at the standard marker *-ja-r* in Tanti, meanwhile, we see that it invariably modifies an oblique stem, like any other chain of case suffixes. The oblique stem marker is sensitive to the lexical class and number of the standard of comparison (24a): *-li* for singular nouns, *-a* for plural nouns, *-i* for pronouns; some nouns do not take an oblique stem marker, such as *xuri* 'rock' in (18a), cf. parallel examples from Tsugni and Tanti in (24).

(24) Tanti Dargwa

a. *hit* *dubur-li-ja-r* / *dubur-t-a-ja-r* / *hiž-i-ja-r*
that(ABS) mountain-OBL-SUPER-EL mountain-PL-OBL-SUPER-EL/ this-OBL-SUPER-EL
haq-le *sa-j.*

⁸ In (20)–(21), we give intended translations of ungrammatical forms.

high-ADV COP-M

‘He is taller than the mountain / than the mountains / than this.’ (Lander, Mamaev 2021)

Tsugni Dargwa

- b. *it dubur-le.r.ka / dubur-te-le.r.ka / ij-le.r.ka aq-le ca-j.*
that(ABS) mountain[-COMP] mountain-PL[-COMP] this[-COMP] high-ADV COP-M
‘He is taller than the mountain / than the mountains / than this.’

Stem variation is a distinctive feature of Adverbial vs. Particle comparatives: comparative particles do not show any variation conditioned by lexical and/or grammatical properties of the standard of comparison. In Tsugni, we could expect stem variation only between the singular and plural forms, because in the singular, the in-relative markers normally attach to direct stems (while all other localizations are derived from oblique stems). In other languages, where the locative markers used for comparison consistently attach to oblique stems, the difference is more evident; we will return to this difference in Section 5.

4.2.2 Comparatives as biclausal structures

An important feature of comparative constructions in Tsugni is revealed in the examples where the comparee is in an oblique (non-absolute) case. Consider example (25), where the comparee (*t.at:i-j* ‘to the father’) is in the dative. In the standard of comparison, *zumra-j-le.r.ka* ‘than to the neighbor’, the standard marker *le.r.ka* does not modify a nominal stem: it attaches to the dative form, mirroring the case of the comparee.

Tsugni Dargwa

- (25) *t.at:i-j zumra-j-le.r.ka ʔaʕχ-le b-alχ-a.*
father-DAT neighbor-DAT[-COMP] good-ADV N-know.IPFV-TH(3)
‘Father knows better than the neighbor.’

The same can be seen in sentences (26)–(27). In (26), the comparee *e-ge-r-ka* ‘about you’ takes the super-relative case, which is repeated in the standard of comparison *dam-ge-r-ka-le.r.ka* ‘than about me’. In (27), the noun representing the standard of comparison (*uškul* ‘school’) includes two successive occurrences of the sequence *lerka* – the first as a locative marker, the second as a standard marker.

Tsugni Dargwa

- (26) *du e-ge-r-ka imc’a-le pikri-w-ik’-u-le=da.*
I(ABS) you.SG.OBL-SUPER-EL-DOWN more-ADV thought-M-LV.IPFV-PRS-CVB=1
dam-ge-r-ka-le.r.ka
I.OBL-SUPER-EL-DOWN[-COMP]
‘I think about you more than about myself.’
- (27) *u uškul-le-r-ka-le.r.ka it ʔaʕči-le-r-ka*
you.SG(ABS) school-LOC-EL-DOWN[-COMP] that(ABS) work-LOC-EL-DOWN
čaʕw-le č:ar-r-iχ-ub.
early-ADV back-F-become.PFV-AOR
‘She got home from work earlier than you [got home] from school.’

Attaching to case-marked nominal forms, the standard marker confirms our assumption that it is an enclitic particle rather than a suffix or a sequence of suffixes.

How can we explain the fact that the case of the standard of comparison repeats the case of the comparee? The most obvious way is to suggest that the standard of comparison belongs to a subordinate clause where it is governed by an elided predicate that lexically coincides with the predicate of the main clause, as shown in (28).

1
2 Tsugni Dargwa
3 (28) *dali imc'a d-erk-un=da q:a'mpit'-e [eli d-erk-un-ce-le.r.ka]*.
4 I.ERG more NPL-eat.PFV-AOR=1 candy-PL(ABS) you.ERG NPL-eat.PFV-AOR-ADJ[-COMP]
5 'I ate more candy than you.'

7 The comparative construction where the standard of comparison belongs to a dependent clause and
8 is marked by a particle is an instance of a Particle comparative in terms of Stassen (1985; 2013). In such
9 constructions, certain parts of the dependent clause are elided so that – in the simplest case – only
10 the standard of comparison itself remains. The standard marker functions as a subordinate
11 conjunction.

12 A variety of biclausal analysis has long been the most common approach to the structure of
13 Particle comparatives; several syntactic models have been suggested (cf., for example, Bresnan 1973;
14 Kennedy 2002; Lechner 2004; Bacsikai-Atkari 2014)⁹.

15 A detailed study of the syntactic properties of the Tsugni comparative lies outside the scope
16 of this paper. Here we are simply trying to show that in Tsugni, comparative constructions are biclausal.
17 The main argument favoring this is the fact that the case of the standard of comparison repeats that
18 of the comparee. This analysis is also supported by the distribution of the standard markers, which is
19 discussed in Section 4.2.3.

20 The elided predicate of a comparative clause can be restored – in a non-finite form – but in
21 that case the comparative particle attaches to the predicate, cf. (29a), where the predicate is elided,
22 and (29b) where it is present and followed by the standard marker. By contrast, sentence (29c), where
23 the predicate is present but the standard marker follows the NP *eli* 'you.ERG', is ungrammatical¹⁰.

24
25 Tsugni Dargwa
26 (29) a. *dali imc'a d-erk-un=da q:a'mpit'-e [eli-le.r.ka]*.
27 I.ERG more NPL-eat.PFV-AOR=1 candy-PL(ABS) you.ERG[-COMP]
28 'I ate more candy [than you].'
29 b. *dali imc'a d-erk-un=da q:a'mpit'-e [eli d-erk-un-ce-le.r.ka]*.
30 I.ERG more NPL-eat.PFV-AOR=1 candy-PL(ABS) you.ERG NPL-eat.PFV-AOR-ADJ[-COMP]
31 'I ate more candy [than you]' (lit. 'than what you ate').
32 c. **dali imc'a d-erk-un=da q:a'mpit'-e [eli-le.r.ka d-erk-un-ce]*.
33 I.ERG more NPL-eat.PFV-AOR=1 candy-PL(ABS) you.ERG[-COMP] NPL-eat.PFV-AOR-ADJ
34 (Intended translation:) 'I ate more candy [than you ate].'

36 Only a biclausal analysis is applicable to the constructions where comparison is based on more than
37 one parameter, illustrated in (30); cf. also (27) above. As in (29), we can either compare two clauses
38 and place the standard marker after the predicate (29a) or omit the predicate and place the marker
39 after the patient NP (29b).

40
41 Tsugni Dargwa
42 (30) a. *asijat-li-j cunna gul-e b-ik:-an-ce-le.r.ha,*
43 PN-OBL-DAT self.GEN children-PL(ABS) HPL-want.IPFV-PTCP-ADJ[-COMP]
44 *pat'imat-li-j cunna kat-ne imc'a-le d-ik:-u.*
45 PN-OBL-DAT self.GEN cat-PL(ABS) more-ADV NPL-want.IPFV-TH(3)
46 'Patimat loves her cats more than Asiyat loves her children.'

⁹ Some instances of Particle comparatives have been analyzed as prepositional phrases (for example, Kennedy 1999 for English) but most constructions are better accommodated by the biclausal analysis.

¹⁰ This makes us suggest that the standard of comparison takes a special position in the dependent clause, which is followed by *the standard marker/conjunction*. In (29a) this position is taken by the NP *eli* 'you.ERG', and in (29b) by the phrase *eli derkunce* 'eaten by you', which is headed by a participle.

b. *asijat-li-j* *cunna* *gul-e-le.r.ha*,
 PN-OBL-DAT self.GEN children-PL[-COMP]
pat'imat-li-j *cunna* *kat-ne* *imc'a-le* *d-ik:-u*.
 PN-OBL-DAT self.GEN cat-PL(ABS) more-ADV NPL-want.IPFV-TH(3)
 'Patimat loves her cats more than Asiyat [loves] her children.'

In example (31) from Tanti, which is synonymous with sentence (26) from Tsugni, the case form of the comparee does not influence the form of the standard of comparison: the comparee appears in the super-essive, while the standard of comparison takes the super-relative case (*damži-ja-r* 'than me'), just as in ABS-comparison (cf. 18a, 24a). This is a clear case of an Adverbial comparative.

Tanti Dargwa
 (31) *du* *ʕaʕži-ja-w* *imc'a-le* *pikri-w-ik'-u-l=da* *damži-ja-r*.
 I(ABS) you.SG.OBL-SUPER-M(ESS) more-ADV thought-M-LV.IPFV-PRS-CVB=1 I.OBL-SUPER-EL
 'I think about you more than about myself.'
 (Lander, Mamaev 2021)

4.2.3 Distribution of the standard markers

Comparison is possible not only between entities that are encoded by nouns or other declinable words¹¹, but also by non-declinable words like adverbs (*Yesterday was colder than today*; *It is more dangerous upstairs than downstairs*) or verb forms (*The child crawls faster than he walks*; *Sitting at home makes me more tired than working all day*). In the East Caucasian languages, the non-declinable words that can be compared are adverbs, converbs, infinitives, and, less frequently, postpositions. When discussing the comparison of non-declinable words, we will speak about *ND-comparison*.

Case suffixes, which are used as standard markers in East Caucasian Adverbial comparatives, cannot attach to non-declinable words; hence, Adverbial comparatives cannot be directly used for ND-comparison. If a standard marker easily combines with words that are otherwise non-declinable, we can conclude that it is a particle, not a case suffix.

In Tsugni, non-locative adverbs, infinitives and converbs cannot combine with case markers, yet, as shown in (32)–(34), they are fully compatible with the standard markers *le.r.ka* and *le.r.ha*.

Tsugni Dargwa
 (32) *s:a* *iberi-le.r.ka* *b-uχ:ar-le=de*.
 yesterday today[-COMP] N-cold-ADV=PST
 'Yesterday was colder than today.'
 (33) *dars* *ka-b-iš:-i-le.r.ka* *duče* *uq'-i* *b-ik:-ar*.
 lesson(ABS) DOWN-N-put.PFV-INF[-COMP] herding (M)go.IPFV-INF N-like.IPFV-TH(3)
 'I'd rather be a shepherd [= go herding] than teach at school.'
 (34) *zumr-up:e* *ak:ʷar-le-le.r.ka* *le-b-le* *ʔaʕχ-ce* *caʕb>i*.
 neighbor-PL(ABS) NEG.COP-CVB[-COMP] EXST-HPL-CVB good-ADJ COP<N>
 'It's better to have neighbors than not to have them.'

Locative adverbs in Tsugni and other Dargwa lects have partial declension: they can take markers of orientation and direction. The standard markers *le.r.ka* and *le.r.ha* attach to entire adverbial forms, which may have their own orientation and direction suffixes. In (35), the presence of a gender marker

¹¹ As mentioned in Section 2, we refer to individual words here as a shorthand for the constituents that they head.

at the end of the adverbial form *q:at:a-b* ‘outside’ marks this form as essive, while in (36) the adverb *salarka* ‘ahead, earlier’ contains the elative suffix *-r* and the directional suffix *-ka* ‘down’.

- Tsugni Dargwa
- (35) *ilt:u-b-le.r.ka* *išt:u-b* *er-w-irχ-i* *ʔaʕχ-le* *cai*.
 there-N(ESS)[-COMP] here-N(ESS) live-(M)LV.PFV-INF good-ADV COP<N>
 ‘Living here is better than there.’
- (36) *sala-r-ka-le.r.ka* *ħanna* *ʔaʕχ-le* *b-ig-u-le* *cai*.
 forward-EL-DOWN[-COMP] now good-ADV N-see.IPFV-PRS-CVB COP<N>
 ‘You can see better now than before’ (locative adverb in temporal meaning).

The only open word class that does not take the standard markers *le.r.ka* and *le.r.ha* are finite verbs, cf. the ungrammatical sentence (37a). If we want to compare two predicates, then one of them, namely the standard of comparison, must appear as a participial head of a dependent clause, as shown in (37b).

- Tsugni Dargwa
- (37) a. **sala-r-ka* *er-d-irχ^w-ub=da-le.r.ka* *ħanna* *ʔaʕχ-le*
 forward-EL-DOWN live-1/2PL-LV.IPFV-PRET=1[-COMP] now good-ADV
er-d-irχ^w-u-l=da.
 live-1/2PL-LV.IPFV-PRS-CVB=1
 Intended translation: ‘We live better now than [we did] before.’ (lit. ‘Than we lived before, we live well now.’)
- b. *[sala-r-ka* *er-d-irχ^w-ub-ce-le.r.ka]* *ħanna* *ʔaʕχ-le*
 forward-EL-DOWN live-1/2PL-LV.IPFV-PRET-ADJ[-COMP] now good-ADV
er-d-irχ^w-u-l=da.
 live-1/2PL-be.IPFV-PRS-CVB=1
 ‘We live better now than [we did] before.’ (lit. ‘Than living before, we live well now.’)

In the previous section we suggested that the standard of comparison belongs to a subordinate comparative clause. The fact that the standard markers cannot attach to finite verbs supports this suggestion, because, as mentioned in Section 3.1, finite forms do not normally head dependent clauses.

The standard markers in Tsugni combine with all open word classes except finite verbs, which is again more typical of particles than of suffixes. If we look at ND-comparison in Tanti, we find that it uses a more complex strategy: the speaker uses the suffix *-il* to derive an adjective from the adverb meaning ‘there’, and then adds the oblique stem marker *-li* and the locative suffixes to the adjective.

- Tanti Dargwa
- (38) *hit:u-b-il-li-ja-r* *hišt:u-b* *ʕer-iχ^w-iž* *ʔaʕχ-le* *sa-b*.
 there-N(ESS)-CONTR.ADJ-OBL-SUPER-EL here-N(ESS) live-(M)LV.PFV-INF good-ADV COP-N
 (a=b) ‘Living here is better than there.’

The strategy of ND-comparison illustrated in (38) is very common in the East Caucasian family. Since the standard of comparison is expected to be in a certain case form, it must be a noun or another declinable word (adjective, numeral, pronoun). This is why Tanti and other languages with Adverbial comparatives use nominalization for ND-comparison. By nominalization we mean the derivation of both nouns (substantivization) and adjectives (adjectivization). If nominalization is accessible for different word classes, then we can easily compare adverbs and verb forms through their

transformation into nouns, adjectives or participles even in a language that only has Adverbial comparatives.

4.3 Comparative particle in Tsugni: a case of degrammaticalization?

In Sections 4.2.1–4.2.3, we showed that the standard markers *le.r.ka* and *le.r.ha* are clitics rather than suffixes. These clitic markers attach to whole word forms marked for number and case. They have a broad distribution and can modify content words of any kind, with the exception of finite verbs. Comparative constructions in Tsugni are biclausal, with the standard marker introducing a comparative dependent clause. This means that we can classify the comparative constructions of Tsugni as Particle comparatives. The Particle comparatives in Tsugni differ from typical European comparatives in that adjectives and adverbs do not have degrees of comparison.

In Tsugni, the Particle comparatives obviously go back to Adverbial comparatives. The particles used are segmentally identical to two trimorphemic markers of the in-relative case. We can also compare Tsugni to other lects of the Dargwic group – in most of them, as in Tanti, the standard is marked by similar locative case markers.

The development from a spatial marker to the comparative particle can be viewed as a case of degrammaticalization (see also Section 6). In Tsugni, this process included morphological merger (*-le-r-ka* > *=lerka*), which is rather typical of grammaticalization.

The question arises as to what factors influenced this development; both internal factors and language contacts could be relevant.

The dominant language in the area is Russian, and it has Particle comparatives (example 5 above). However, we are hesitant to attribute the unusual comparatives of Tsugni to Russian influence. The village of Tsugni (Map 1) is located in the south-west of the Dargwa area, where most villages speak their own individual lect. There are several lects that are relatively close to Tsugni genetically and geographically and have been described, including Itsari (Sumbatova & Mutalov 2003), Sanzhi (Forker 2020), and Tanti (Sumbatova & Lander 2014), and to our knowledge all these lects retain the original Adverbial comparatives. Tsugni has not been exposed to Russian influence more than any of these lects. The syntax of Tsugni, as well as that of other small lects in the region, generally maintains its structure and shows minimal Russian influence, making it unlikely that Russian affected a single construction in a single lect in such a selective manner. Besides, the Russian comparative particle *chem* has nothing in common with locative case forms (it originates from the instrumental case of the pronoun *что* ‘what’).

An alternative possibility is the influence of another contact language: for Tsugni, a reasonable candidate would be Lak. Lak constitutes a separate group within the East Caucasian family, and is often believed to be the closest relative of Dargwa.

The inhabitants of Tsugni have had close contacts with the Lak people. The Lak villages of Kumukh, Vachi, Kuli, Ulluchara and others are relatively close to Tsugni (Map 1). Marriages have taken place between the Laks and the speakers of Tsugni; the people of Tsugni used to meet Lak speakers when visiting important markets in the area, one of them in the large Lak village of Kumukh; and in Soviet times, the children of both Dargwa and Lak people were sometimes brought up together in boarding schools. As a result of these contacts, there have been a number of Tsugni–Lak bilinguals. So, some contact phenomena are quite possible, and it would make sense to look at comparative constructions in Lak.

Lak (Kazenin 2013: 287–292; Friedman 2020: 210–212, 217) has two comparative constructions: an Adverbial construction with a dedicated case suffix *-jar* and a Particle comparative with the particle *=nijar*, which supposedly originates from the original comparative suffix *-jar* (or, to be more exact, from the combination of *-jar* with the oblique stem marker *-ni*; for more details, see Section 5.5). The situation in Tsugni is to a certain degree similar – as we suggest, the modern comparative particles go back to trimorphemic relative markers – but the details are different. In Lak, the new standard marker *=nijar* did not oust the original suffix *-jar* but coexists with it. Morphologically, the standard markers in Lak and Tsugni are not exactly parallel either.

Lak may have influenced the development of comparative constructions in Tsugni, but we think that the latter has been supported by internal factors – in particular, by the analogy with equative constructions (comparatives of equality). The equative constructions are formed with the help of identity particles *=q'a^cjda* and *=da^cq'/=da^cq'le* (in free variation) that attach to the standard of comparison; cf. examples (39b) and (40b). Syntactically, equative constructions in Tsugni are parallel to comparatives and differ from them only in the choice of the particle. For any comparative construction, there exists a parallel equative construction, differing only in terms of the standard marker; cf. (39a) and (39b), (40a) and (40b).

- Tsugni Dargwa
- (39) a. *s:a iberi=lerka b-uχ:ar-le=de.*
 yesterday today=COMP N-cold-ADV=PST
 'Yesterday was colder than today.'
- b. *s:a iberi=q'a^cjda b-uχ:ar-le=de.*
 yesterday today=EQUAT N-cold-ADV=PST
 'Yesterday was as cold as today.'
- (40) a. *ʒurus mec-li=lerka niš:ala mec-li q:uwa-le*
 Russian language-ERG=COMP we.GEN language-ERG beautiful-ADV
ʋaj-b-ik'^w-ar.
 word-HPL-LV.IPFV-TH(3)
 'They speak our language better than [they speak] Russian.'
- b. *ʒurus mec-li=q'a^cjda niš:ala mec-li q:uwa-le*
 Russian language-ERG=EQUAT we.GEN language-ERG beautiful-ADV
ʋaj-b-ik'^w-ar.
 word-HPL-LV.IPFV-TH(3)
 'They speak our language as well as [they speak] Russian.'

Influence from the equative constructions on the comparatives is possible, but this does not seem to be an influential factor either. In other lects of the Dargwic group, the equative construction is structured similarly to that seen in Tsugni, but this has not caused any changes in the structure of the comparatives. We think that another internal factor is more relevant in the development of a Particle comparative in Tsugni, namely, a general tendency towards the emergence of Particle comparatives in the East Caucasian languages. We will discuss this idea in the next section.

5 Comparative constructions across the East Caucasian languages: a pilot study

5.1 Research questions

In Section 5, we describe comparative constructions in different languages of the East Caucasian language family. For each language, we attempt to determine whether it has adverbial or/and Particle comparative constructions. For the languages with Adverbial constructions, we check if the standard of comparison is expressed by a locative or a dedicated case form. If a language has a Particle comparative, we check whether it goes back to a previous Adverbial comparative or to an alternative source. For languages without Particle comparatives, we look at the strategy they use for ND-comparison.

We are often forced to identify the construction type on the basis of relatively scarce data. To do that, we consistently look for the three classes of comparative sentences we described in Section 4.2 – ABS-comparison (the comparee is an absolutive NP), OBL-comparison (the comparee is an NP in an oblique case, and ND-comparison (the comparee is headed by a non-declinable word).

In ABS-comparative sentences, we check whether the standard of comparison shows stem variation. We know that most languages of the East Caucasian family display two-stem declension, whereby most non-absolutive case forms are derived from a distinct oblique stem. Some languages have several types of oblique stem formation, distributed lexically and/or grammatically. In an Adverbial comparative, the standard of comparison appears in a particular case form, usually derived from an oblique stem. So, on the standard of comparison, oblique stem variation shows that comparison is encoded by case markers, which is an argument in favor of the Adverbial comparative. If there is no such variation where it is expected, this could mean that we are dealing with an independent element, probably a particle.

Looking at OBL-comparison is the most direct way to determine whether a comparative construction is biclausal or not. If we see that the standard marker attaches to a full case-marked form of a noun, and the case of the standard of comparison repeats the case of the comparee, then we are dealing with a biclausal Particle comparative.

Examples of ND-comparison show the distribution of the standard marker: if it is compatible with non-declinable words without any additional marking, then it should be treated as a particle. We also look at the strategies that are used for ND-comparison: languages with Adverbial comparatives tend to use nominalization.

The initial results of our study are shown in Table 2. The left header of the table shows whether the language has an Adverbial comparative. The top header contrasts languages with and without Particle comparatives. Thus, the four cells of the table correspond to three groups of languages: (1) languages with Adverbial comparatives only, (2) languages with Particle comparatives only, (3) languages with both types of comparative constructions; the fourth cell, which would correspond to languages without either of the two constructions, is empty.

Table 2. Adverbial and Particle comparatives in the East Caucasian languages¹²

		PARTICLE COMPARATIVES	
		no	yes
ADVERBIAL COMPARATIVES	no		Kajtag, Tabasaran, Tsugni Dargwa
	yes	Avar, Ingush, Hinuq, Hunzib, Khinalug, Lezgian, Sanzhi Dargwa, Tanti Dargwa, Tsakhur, Tsez (Andi, Batsbi, Chechen, Rutul, Tindi, Kryz)	Agul, Akhvakh, Archi, Bagvalal, Godoberi, Karata, Khwarshi, Lak, Literary Dargwa, Udi (Chamalal, Bezhta, Botlikh, Budukh)

As seen from Table 2, Tsugni is far from unique in having Particle comparatives. In Sections 5.2–5.5, we will show that there is also variation within the main groups as to the origin and internal structure of the standard markers: in most languages they go back to spatial markers, while in others they come from an alternative source.

5.2 Languages with Adverbial comparatives only

Languages that possess only Adverbial comparatives constitute one of the biggest groups in the East Caucasian family. In most of these languages, the standard of comparison in such constructions is encoded by a locative case form (for details and variation see Naccarato 2020).

For example, in Lezgian (Haspelmath 1993: 432–434), the standard of comparison is in the super-relative case with the marker *-laj*. This marker attaches to an oblique stem, which is formed differently for different nouns and pronouns: for example, with oblique marker *-di* in *xala-di-laj* ‘than aunt’ (41), *-č’e* in *č’ar-č’e-laj* ‘than hair’ (42), and no oblique marker in *ne-laj* ‘than anyone’ (43). Example (41) also shows that the case of the standard does not repeat the case of the comparee: the comparee is in the ergative, whereas the standard of comparison is in the super-relative case.

¹² If the construction type is identified only tentatively at this point, the language name is given in brackets.

Lezgian
 (41) *Dide-di tort-ar xala-di-laj ijer-diz čra-da.*
 mother-OBL(ERG) cake-PL aunt-OBL-SUPER.EL good-ADV bake-FUT
 ‘Mother bakes cakes better than aunt.’ (Haspelmath 1993: 433).

(42) *Eger za tab-zawa-t'a, zi gardan wa-z č'ar-č'e-laj*
 [if I:ERG lie-IPFV-COND] I:GEN neck you-DAT hair-OBL-SUPER.EL
 čük'ü xu-j.
 thin be-OPT
 ‘If I am lying, may my neck be thinner than a hair for you.’ (Haspelmath 1993: 395).

(43) *Jusuf-a ne-laj xajit'ani q^hsan-diz mani-jar luhu-zwa.*
 Jusuf-OBL(ERG) who-SUPER.EL INDEF good-ADV song-PL say-IPFV
 ‘Jusuf sings better than anyone.’ (Haspelmath 1993: 196).

Some languages in this group have dedicated comparative case suffixes. However, their comparative constructions also belong to the Adverbial type, since these constructions meet the criteria listed in Section 3.3: they are monoclausal, the standard marker is a nominal suffix, and the standard of comparison is an adverbial phrase. For example, Ingush (Nakh) has a dedicated comparative case with the suffix *-l*, in plural *-el* (Nichols 2011: 127; 425–427; 516–517).

Ingush
 (44) *Xii benziinal dazagh dy.*
 water gasoline.COMP D.heavy.CMP D.be.PRS
 ‘Water is heavier than gasoline.’ (Nichols 2011: 516)

For ND-comparison, both Lezgian and Ingush use nominalizations. In sentences (45)–(46) from Lezgian, the substantivizer *-da* is used. In (45), the temporal adverb *gilan*, which functions as the standard of comparison, is nominalized as a relative adjective (with the suffix *-da*) and takes the super-relative suffix *-laj*.

Lezgian
 (45) *Lezgi č'al-a inlaj wilik padež-r.i-n q'adar*
 Lezgian language-IN.ESS hither before case-PL-GEN number
gilan-da-laj artux tir.
 present-SBST-SUPER.EL more COP:PST
 ‘The number of cases in Lezgian was formerly greater than now.’ (Haspelmath 1993: 433)

Example (46) shows the postpositional phrase *t'ur galačiz* ‘without a spoon’ as the standard of comparison, with the nominalization of the deverbal postposition *galačiz* ‘without’.

(46) *T'ur gala-č-ir-da-laj t'ur.una-ldi šurpat ne-z rehät ja.*
 [spoon be.with-NEG-PTCP-SBST-SUPER.EL spoon-SUPER.DIR soup eat] easy COP
 ‘It is easier to eat the soup with a spoon than without a spoon.’ (Haspelmath 1993: 433)

Like Lezgian, Ingush uses nominalizations when comparing non-declinable parts of speech; what is more, it nominalizes oblique case forms in OBL-comparison. Johanna Nichols (2011: 161) explains that “[g]enitive case forms of nouns and pronouns, ablatives, and various adverbial forms can add nominalizing morphology with the semi-periphrastic declension <...>, in which the nominative adds the nominalized form of ‘be’ and the oblique cases take the extension *-chu-*”. These nominalizations

have a range of different uses; in particular, they can transform a non-declinable word into a noun if it is needed as a standard of comparison. Like all nouns, nominalizations take the comparative case suffix, cf. (47) with a nominalized dative form of the pronoun ‘I’, (48) with the adverb ‘now’ and (49) with a participle.

- Ingush
- (47) *...hwuona suonachyl duqqa=’a duqagh xazaaxuddy yz.*
 you.SG.DAT I.DAT.NMLZ.COMP much:FOC=ADD more hear-INFR.D it
 ‘You must have heard much more about it than I have.’ (Nichols 2011: 425)
- (48) *yz cu xaanara +dar=’a dikagh daga+doagha handzchyl.*
 DEM DEM.OBL time.EL +D.NMLZ=ADD good.CMP remember+D.LV.PRS now.NMLZ.OBL.COMP
 ‘You remember things from those days better than today’s.’ (Nichols 2011: 163)
- (49) *uqaza kyljgagh dulachul aattach dy*
 here hand.PL D.wash.PTCP.NMLZ.COMP easy.CMP D.be.PRS
vaina dehwa=’a deanna.
 we.INCL.DAT across=ADD D.go.CVant
 ‘It will be easier for us to go over to the other side (of the lake) than wash our hands here.’
 (Nichols 2011: 291)

Nominalization as a strategy for ND-comparison is also found in Avar, Hinuq, Sanzhi Dargwa, Tsakhur, Tsez and other languages (see Table 3 in Section 6 below).

5.3 Languages with Particle comparatives only

A relatively small group of East Caucasian languages have only Particle comparatives (Tsugni Dargwa, Tabasaran, Kajtag).

Tsugni Dargwa, discussed in Section 4, is a language where the Particle comparative obviously goes back to an Adverbial comparative.

Tabasaran (Lezgic) also has only Particle comparatives with the standard marker =*t’an* (Magometov 1965: 335). This particle is used in all classes of comparative constructions and attaches to words belonging to various parts of speech, including nouns in different cases, adverbs and non-finite verb forms (examples 50–52). The case of the standard of comparison repeats the case of the comparee (51). The particle =*t’an* includes the elative marker *-an*, but if we accept that the particle =*t’an* originates from an elative marker, the origin of the initial *t’* remains unclear. Another possible hypothesis is that was borrowed from Azerbaijani (Turkic). The Azerbaijani ablative case suffix *-dan* also functions as standard marker.

- Tabasaran
- (50) *yarxla? a-yi čve=t’an bagah xha-yi ğunši užu vu.*
 in.the.distance be.in-PTCP brother=than near be.near-PTCP neighbor good COP
 ‘A neighbor who is near is preferable to a brother who is far away.’ (Babaliyeva 2013: 136).
- (51) *yarkvrrar-i-?t’an jvuv.a-n bagah uju-di vu.*
 forest.PL-OBL-IN=than self-GEN near good-man COP
 ‘Better close to home than in the woods.’ (Babaliyeva p.c.).
- (52) *ux-ž:i=č’an k’an-ž:i=ra xh-ub už:uvu.*
 quick-ADV=than late-ADV=ADD be-MSD good
 ‘Better late than never.’ (Genko 2005: 15)

Particle comparatives are also found in Kajtag (Dargwic). Unfortunately, there is no description of these constructions, but there are examples showing that Kajtag has a Particle comparative with the particle *at:en / et:an / et:en* (Temirbulatova 2004; Pavel Dolgoplov p.c.). Most probably, this marker or a part of it was borrowed from Kumyk (Turkic), where the standard of comparison is marked by the ablative suffix *-dan* (Abdullaev et al. 2014).

- Kajtag (Dargwa)
- (53) *d-aqil* *ɤaj* *et:an* *ɤa-il* *čut:u* *ʔaʕh-ti* *d-irh^w-ar.*
 NPL-much word(ABS) than twenty-CARD pie(ABS) good-ADJ.PL NPL-be.IPFV-TH(3)
 ‘Twenty pies are better than a lot of talk.’ (Temirbulatova 2002: 264)
- (54) *dila* *uba-l* *s:uk:e-l* *et:en* *ʔaʕh-li* *bukan* *b-ir-ar.*
 I.GEN mother-ERG all-ERG than good-ADV food(ABS) N-make.IPFV-TH(3)
 ‘My mom cooks better than anyone.’ (Pavel Dolgoplov, p.c.)
- (55) *ʔaʕmru-l-c:i* *ha-r-eɤ-ib-il* *rirsi* *ħir-r-ar-ara* *et:an*
 life-OBL-INTER UP-F-reach.PFV-AOR-PTCP girl(ABS) live-F-make.PFV-INF than
ajɤur *irči* *ħir-b-ar-ara* *samat-li* *b-irh^w-ar.*
 stallion horse(ABS) live-N-make.PFV-INF light-ADV N-be.IPFV-GEN.PRS.3
 ‘A stallion is easier to keep than [to keep] an adult daughter.’ (Temirbulatova 2004: 263)

5.4 Languages with two unrelated comparative constructions

Agul, Godoberi, Karata, Udi and probably Botlikh have Adverbial comparatives and Particle comparatives that are not diachronically related.

Agul (Lezgetic) has an Adverbial (Locational) comparative where the standard of comparison is in the super-relative case marked by *-la-s* and a Particle comparative with the particle *=dala* (Maisak 2008; Maisak 2014: 423; Magometov 1970: 84). The particle is possible in all classes of comparatives (56b, 57b, 58b), while the case marker attaches to nouns in both ABS-constructions (56a) and OBL-constructions (57a).

- Agul
- (56) a. *we* *χul-a-l-as* *ze* *χal* *bat'ar-f* *e.*
 your(SG) house-OBL-SUPER-EL my house beautiful-SBST COP
 (Maisak 2008)
- b. *we* *χal=dala* *ze* *χal* *bat'ar-f* *e.*
 your(SG) house=COMP my house beautiful-SBST COP
 ‘My house is more beautiful than yours.’ (Merdanova, Maisak p.c.)
- (57) a. *gada-ji-s* *p:ara* *q:enfet-ar* *i-ne* *ruš-a-la-s.*
 boy-OBL-DAT many candy-PL give.PFV-AOR girl-OBL-SUPER-EL
- b. *gada-ji-s* *p:ara* *q:enfet-ar* *i-ne* *ruš-a-s=dala.*
 boy-OBL-DAT many candy-PL give.PFV-AOR girl-OBL-DAT=COMP
 ‘The boy was given more candy than the girl.’ (Merdanova, Maisak p.c.)
- (58) *χup:aj* *me-bur* *me* *kasib-di,* *“za-s* *arg^wa-j*
 then this-PL this poor-OBL(ERG) I-DAT see.IPFV-CVB
lūk'ü-j=dala, *za-s* *d-arg^wa-j* *k'i-na*
 die.IPFV-CVB=COMP I-DAT NEG-see.IPFV-CVB die.PFV-CVB
iži *e,”* *pu-na* *fātu-ne.*
 good COP say.PFV-CVB APUD.EL.let.PFV-AOR

1 'Then the poor man said: "Let them die so that I may not see it, rather than dying before my
2 eyes," and let them go.' (Maisak 2014: 94–95)

3
4 Historically, the particle =*dala* has no connection with the case marker *-la-s*. Its primary meaning is
5 'with the exception of' (59).

6
7 Agul
8 (59) *mi-q düja-ji-l sa tufang=na sa geda=dala fi=ra*
9 this-POST world-OBL-SUPER one gun=and one son=COMP what=ADD
10 *qa-dawa-j xu-ni.*
11 POST.be-NEG-CVB become.PFV-AOR
12 'He had nothing in the world but his gun and his son.' (Maisak 2014: 293)

13
14 A similar situation is observed in Godoberi (Sosenskaja, Tatevosov 1996: 167–169): Godoberi has
15 Adverbial comparatives with the cont-relative suffix *-č'u-ru*, and Particle comparatives with marker
16 *tikuru* (with nouns and adverbs) or *-qu* (with verb forms). The origin of these particles is unknown, but
17 the element *ru* within *tikuru* may be a trace of the former relative suffix.

18 In Udi (Lezgif; Maisak 2018) the standard of comparison is usually marked by the ablative
19 (relative) suffix *-(a)χun* (Adverbial comparative), which attaches to nouns in different cases and, rarely,
20 to infinitives; there is also an alternative construction that is used when the standard of comparison is
21 a clause, which uses the marker *-amin* originally meaning 'until, by the time' (Particle comparative).

22 One more language with two comparative constructions is Archi (Lezgif). It has an Adverbial
23 comparative with the dedicated case marker *-χur* and a Particle comparative with the particle *kilaw*
24 (Chumakina p.c.). With nouns, both constructions are possible (60a, 61a), but the *kilaw*-construction
25 seems to be more natural when the comparee is non-absolutive (OBL-comparison), as in (61b), and it
26 is the only possible construction in ND-comparison (62).

27
28 Archi
29 (60) a. *dogi niʃi-χur t'i.*
30 donkey(IV)[ABS.SG] horse(III).OBL.SG-COMP be.small
31 b. *dogi noʃ kilaw t'i.*
32 donkey(IV)[ABS.SG] horse(III)[ABS.SG] than be.small
33 (a=b) 'A donkey is smaller than a horse.' (Chumakina p.c.)
34
35 (61) a. *Moħommal-li-s labχanši guručammul kto-li*
36 Magomed-OBL.SG-DAT a.lot sweet(IV)-PL.ABS [IV.PL]give.PFV-EVID
37 *Rasul-li-χur.*
38 Rasul-OBL.SG-COMP
39 b. *Moħommal-li-s labχanši guručammul kto-li*
40 Magomed-OBL.SG-DAT a.lot sweet(IV)-PL.ABS [IV.PL]give.PFV-EVID
41 *Rasul-li-s kilaw.*
42 Rasul-OBL.SG-DAT than
43 (a=b) 'Magomed was given more candies than Rasul.' (Chumakina p.c.)

44
45 (62) *lo χilibχʰi=jχbu bu-kan-ši eχdi*
46 child(II)[SG.ABS] porridge(III)[SG.ABS]=EMPH<III.SG> III.SG-eat.IPFV-CVB <II.SG>be.PST
47 *akt' kummu-s kilaw.*
48 meat(IV)[SG.ABS] [IV.SG]eat.IPFV-FIN than
49 'The girl was eating the porridge, she (likes it better) than eating meat.' (Chumakina & Bond
50 2016: 102)

In Archi, neither of the two comparative markers expresses a spatial meaning¹³, and there is no diachronic relationship between them.

5.5 Languages with two historically related comparative constructions

The most complicated case concerns languages with two comparative constructions where the Particle construction goes back to the Adverbial construction, and both constructions coexist in the modern language. In the East Caucasian family, we can observe different stages of such development.

A relatively early stage is represented by Tanti Dargwa. In Section 4.2, we compared this to Tsugni as a language that has no Particle comparatives. When faced with ND-comparison, it normally uses nominalizations (cf. example 38), but sporadically it adds the spatial case marker directly to an adverb (63). We do not list Tanti and other languages of this type as languages with Particle comparatives, but it is possible that they have started developing in this direction.

- Tanti Dargwa
- (63) *s:a-li-ja-r* *ižberi* *b-uχ:ar-le* *sa-b.*
yesterday-OBL-SUPER-EL today N-cold-ADV COP-N
‘It’s colder today than yesterday.’

A later stage can be found in Khwarshi (Tsezic). This language has an Adverbial comparative where the standard is marked by the super-relative case (Khalilova 2009: 285–291). In most nouns, the super-relative marker follows the oblique stem, which is highly irregular (Khalilova 2009: 53–66).

- Khwarshi
- (64) *idu* *uže* *kandi-λ'o-zi* *toxa-w* *goli.*
this boy(I) girl.OBL-SUPER-EL lazy-I be.PRS
‘This boy is lazier than the girl.’ (Khalilova 2009: 287)

In the comparison of NPs, Khwarshi demonstrates an interesting opposition of “core” cases (absolutive, ergative, experiencer lative), on the one hand, and all other cases, on the other hand. If the comparee is in one of the core cases, the standard of comparison takes the super-relative case, and the case marker *-λ'o-zi* attaches to the oblique stem. If the comparee is in a non-core case, then the same standard marker attaches to a case-marked nominal form.

For example, in both (65) and (66), the comparee is in the lative case, but syntactically the two lative NPs are essentially different. In (65), the comparee is a subject-like experiencer in the lative case; the standard marker *-λ'o-zi* attaches to the oblique stem *isu* of the demonstrative pronoun *žu*. In (66), the lative NP encodes an indirect object; the marker *-λ'o-zi* attaches to the pronoun ‘you’ that is already marked for the lative case.

- Khwarshi
- (65) *isu-λ'o-zi=n* *ø-iže* *geq* *diʃ* *is.*
that.OBL-SUPER-EL=and I-more love.GNT I.LAT sibling(I)
‘I love (my) brother more than he [loves him].’¹⁴ (Khalilova 2009: 291).

- (66) *dubul=λ'o-zi=n* *b-iže* *iλ-i* *de* *ite-l* *os.*
you.SG.LAT=and III-more give-PST.W I.ERG that.OBL-LAT money(III)
‘I gave her more money than [I gave] you.’ (Khalilova 2009: 290).

¹³ Daniel and Ganenkov (2009: 674) note that the Archi comparative *-χur* “is peripherally used to designate spatial adjacency”.

¹⁴ This sentence can also mean ‘I love (my) brother more than [I love] him.’ In this interpretation, the comparee is the absolutive NP.

In other words, in ABS-comparison and partly in OBL-comparison, *-λ'o-zi* behaves as a chain of two suffixes. In most instances of OBL-comparison, *=λ'o-zi* functions as a particle.

Example (67) shows that the particle *=λ'o-zi* can also be added to adverbs and therefore has a broader distribution than case markers do.

- Khwarshi
- (67) *huniža žequt=λ'o-zi l-uc'c'o l-eč-i.*
 yesterday today=UPER.EL IV-cold IV-be-PST.W
 'Yesterday it was colder than today.' (Khalilova 2009: 287)

Khwarshi has developed a new Particle comparative, where the comparative particle *=λ'o-zi* segmentally coincides with the former super-relative marker. However, the Particle comparative is not universal: it is impossible if the comparee is an NP with high syntactic status (absolutive, ergative and experiencer dative).

In Akhvakh (Andic; Creissels 2009), the comparative particle, which also goes back to a spatial marker, has a broader distribution than in Khwarshi. The standard of comparison is marked by the relative case of the general (default) localization *-g-u(ne)*:

- Akhvakh
- (68) *imaḡa-La qāL':ano-g-u reš:ada čula g-o-L-e.*
 donkey.OBL-DAT donkey.foal.OBL-LOC-EL beautiful thing COP-N-NEG-N
 'For the donkey there's nothing more beautiful than its foal.' (Creissels 2009: 19)

Creissels (2009: 27–28) explicitly states that the marker *-g-u(ne)* functions either as a chain of suffixes (a complex case marker) or as a particle. In the first case, the standard marker is added to the oblique stem. The oblique stem marker depends on the gender of the noun (*-s:u* in for masculine nouns, *-t:i* for other nouns); for many nouns, the formation of the oblique stem is irregular ('donkey foal' in 68).

In OBL-comparison and ND-comparison, the marker *=t:une¹⁵*, which is a variant of *-g-u(ne)* taken together with the oblique stem suffix *-t:i*, functions as a particle and attaches to case-marked forms of nouns and to words of other classes, as in (69)–(71). Creissels (2009: 27–28) emphasizes the fact that when functioning as a particle, *=t:une* "is not subject to the selectional restrictions that characterize case endings".

- Akhvakh
- (69) *bešanoč'e āL-e-t:une ič'i harig^w-ada-be šoda-be.*
 a.hundred.times hear-MSD-OBL-(LOC)EL more see-PFV-N good-N
 'Seeing once is better than hearing a hundred of times.' (Creissels 2009: 27)

- (70) *ḡukumate:-λ:i baha baza-λ:i=t:i.g.une iḡ'ae g-o-di.*
 government-GEN price market-GEN=COMP big<N> COP-N-POS
 'Government prices are higher than market prices.' (Magomedova & Abdullaeva 2007: 415)

- (71) *hu-d-e=t:une ha-g-e šoda g-o-di.*
 DIST-SL-ESS=COMP PROX-LL-LOC good COP-N-POS
 'One feels better here than there.' (Creissels 2009: 27)

Akhvakh also developed Particle comparatives from Adverbial constructions, but, unlike in Tsugni and Khwarshi, the comparative particle in Akhvakh includes the former oblique stem suffix. This particle is

¹⁵ According to Creissels (2009), the suffix *-g* is regularly absent in the presence of the oblique stem marker *-t:i*; *-t:une* is a variant of *-t:i-g-une*.

used for OBL-comparison and ND-comparison; in the most common sentences with ABS-comparison, Akhvakh still uses a chain of spatial suffixes.

The final stage of this development can be observed in Lak. In Section 4.3, we mentioned that Lak has an Adverbial comparative with the dedicated suffix *-jar* and a Particle comparative with the marker *=nijar*. As described by Victor Friedman (2020: 210–212), *-jar* functions as a normal case marker and attaches to oblique nominal stems (72a); by contrast, *=nijar* attaches to case-marked nominal forms. If the comparee is an absolutive NP, this form coincides with the direct nominal stem (72b).

- Lak
- (72) a. *us:u s:i-jar qin-s:a Ø-u-r.*
 brother sister.OBL-COMP good-NREF I-COP.PRS-3
 ‘Brother is better than sister.’
- b. *us:u s:u=nijar qin-s:a Ø-u-r.*
 brother sister.ABS=COMP good-NREF I-COP.PRS-3
 ‘Brother is better than sister.’ (Friedman 2020: 217)

When the comparee is a non-absolutive noun, the preferable comparative construction is that with *=nijar*. In this construction, the standard of comparison repeats the case of the comparee (dative in 73). The *jar*-construction is possible, but less common (74).

- Lak
- (73) *žip-ru t'urča wi-č'a=nijar t:u-č'a č'aʁru-r.*
 pocket-PL as.for you.SG.OBL-AD(ESS)=COMP I.OBL-AD:ESS more-COP(3)
 ‘And I’ve got more pockets than you.’ (Kazenin 2013: 289–290)
- (74) *munixlunu x:aj ija cajmi dojarka-qa-jar*
 therefore seem:CVB COP.IPFV other milkmaid-PL-COMP
uza-l nak' č'aʁru-s:a t:izlazi-s:a.
 PN-ERG milk much-ATTR milk.DUR-PTCP.PRS
 ‘Therefore, he thought that Uza (ergative) milked more milk than the other milkmaids.’
 (Kazenin 2013: 288)

Particle constructions are possible in all cases of comparison, including ABS-comparison (72b). Apart from nouns, *=nijar* attaches to words of different classes: pronouns, adverbs (75), postpositions, and non-finite verb forms (76), which is impossible for *-jar*.

- Lak
- (75) *hawa lax:u=nijar veli-nu b-ija.*
 weather yesterday=COMP heat-ADV IV-COP.PST(3)
 ‘The air was warmer than yesterday.’ (Kazenin 2013: 290)
- (76) *maz valba t'i-j=nijar č'aʁwumur č'umal b-ihalawg-un*
 language speech say/LV-CNV=COMP often sometimes III-rest-INF
b-iwt-un qin-s:a=r.
 III-leave-INF good-ATR=COP(3)
 ‘It is often better to let the tongue rest than to talk.’ (Kazenin 2013: 291)

The origin of the comparative particle *=nijar* is not clear in detail. It obviously comes from the combination of an element *ni* with the comparative case marker *-jar*. The first part of this combination, *ni*, may go back to the oblique stem marker *-ni*. Lak has about 30 different oblique stem markers (Friedman 2020: 208–210), and *-ni* is not among the most common of them, if we limit ourselves to

the declension of nouns. At the same time, *-ni* derives oblique stems in several open word classes (demonstrative pronouns, adjectives, participles, numerals) as well as in some important groups of nouns, including personal nouns and ethnonyms (Abdullaev 2007: 45). We therefore find it quite probable that this oblique suffix merged with the comparative case suffix and the new marker *=nijar* expanded over various word classes, including non-declinable words.

A similar situation is found in Literary Dargwa. It has a comparative particle *=iči-GM* (Abdullaev 1961: 186–187; Abdullaev et al. 2014: 213, 357), which clearly originates from the combination of the pronominal oblique suffix *-i* and the super-essive marker *-či-GM*. However, for ABS-comparison, speakers still can use the original super-essive case, which is derived from an oblique stem (80).

- Literary Dargwa
- (77) *q'aʎl niʔ-li-zi-b=iči-b gamuš-la niʔ-li-zi-b ʔaʁwadeš*
 cow milk-OBL-INTER-N=than-N buffalo-GEN milk-OBL-INTER-N fat(ABS)
imc'a-li b-ir-ar.
 more-ADV N-be.IPFV-TH(3)
 'Buffalo milk has a higher fat content than cow's milk.' (Yusupov 2017: 314)
- (78) *abdāl duxu-w-ar-es=iči-b,*
 fool(ABS) smart-M-make.PFV-INF=than-N
emħe duxu-b-ar-es hamad-li sai.
 donkey(ABS) smart-N-make.PFV-INF light-ADV COP<N>
 'A donkey is easier to teach than a fool.' (Yusupov 2017: 383)
- (79) *unc=iči-b q'aʎl aħ-si sai.*
 ox(ABS)=than-N cow(ABS) good-ADJ COP<N>
 'A cow is better than an ox.' (Abdullaev 1961: 187)
- (80) *d-aqal ʁaj-li-či-r ʁal čudu ʔaʁħ-ti d-ir-ar.*
 NPL-much word-OBL-SUPER-NPL twenty pie(ABS) good-PL NPL-be.IPFV-TH(3)
 'Twenty pies are better than a lot of talk.' (Yusupov 2017: 239)

Finally, we wish to mention the very interesting case of Bagvalal (Sosenskaja 2001a). This language has Adverbial constructions employing three different means of standard marking: the super-relative case (*-la-s:*), the cont-essive case (*-č'*) and the affective case (*-ba*) with the postposition *dand* 'against'. The Adverbial constructions are possible with all declinable standards of comparison (ABS- and OBL-comparison). At the same time, Bagvalal also has two comparative particles, *-t:ič'* and *-t:as:*, which go back to the combination of the oblique stem suffix *-t:i* with case suffixes *-č'* and *-la-s:* respectively (Sosenskaja 2001a: 412). As in Lak, the particles are possible with all standards of comparison, including nouns in different case forms and non-declinable words.

- Bagvalal
- (81) *roχ alak-i-la-s: / alak-i-č' / alak-i-ba dand č'ihī ek'a.*
 forest field-OBL-SUPER-EL/ field-OBL-CONT / [field-OBL-AFF against] above is
 'The forest is located higher than the field.' (Sosenskaja 2001a: 410)
- (82) *berozā b-eχa:la-b ek'a sasna=ič' / sasna=as:*
 birch(ABS) N-high-N is pine(ABS)=than / pine(ABS)=than
 'The birch is taller than the pine.' (Sosenskaja 2001a: 412)
- (83) *jaš-li-r waša-š:u-r=ič' / waša-š:u-r=t:as: bešila bihi-r-o:-b keč'.*
 girl-OBL-ERG boy-OBL-ERG=than / boy-OBL-ERG=than well take-IPF-PART-N song
 'The girl sings the song better than the boy.' (Sosenskaja 2001a: 412)

(84) *L'or-a-la-s:* *č'ihis:L'i* *w-a:la* *papa-b* *ek'a.*
rock-OBL-SUPER-EL downward M-come.POT.INF light-N is
hiL'i-s: *č'ih* *w-el-a:=t.as:* / *=t.ič'*
[below-EL upward M-go-MS.POT.INF=than / =than]
'Going down the mountain is easier than going up.' (Sosenskaja 2001a: 412)

The languages discussed in this section (Akhvakh, Bagvalal and others) do not need to use nominalizations for ND-comparison. They compare non-declinable words by expanding the distribution of their usual standard marker, initially a chain of two spatial suffixes (with or without an oblique stem suffix) and attaching it to words like adverbs, infinitives or converbs. This process normally results in developing a Particle comparative.

At the first stage of this development, we have a case marker that attaches to a nominal stem (usually an oblique stem) and consists of two or, less frequently, three suffixes with spatial meaning: (-oblique)-localization-orientation(-direction)

In comparative constructions, the initial spatial meanings of these suffixes are not relevant, and the complex structure of the marker is not supported by other spatial forms. That is why the initial sequence of several suffixes can be perceived as a single marker pointing at the standard of comparison. This single marker begins to be used for ND-comparison. When such usage becomes regular, the former chain of nominal suffixes becomes a particle.

In the languages described in this section we have observed different stages of this process. In Khwarshi, the standard marker combines with case-marked nouns, and the case of the standard repeats the case of the comparee, which is a clear sign of a Particle comparative. But the Particle comparative is still impossible with the most frequent comparees (nouns in the absolutive, ergative and experiencer lative): in these constructions, the standard of comparison is marked by spatial suffixes that in many cases modify an oblique stem. In Akhvakh, the new comparative particle has expanded to all instances of OBL-comparison. Finally, the particle becomes possible with all standards of comparison. This is what we observe in Lak, Literary Dargwa and Bagvalal.

In our data, we have two variants of the development of a comparative particle: the former case marker either transforms into a comparative particle just as it is (Khwarshi), or it merges with the preceding oblique stem marker (Akhvakh, Literary Dargwa, Lak, Bagvalal).

When oblique stem marking becomes part of a comparative particle, it undergoes unification: only one marker is selected from several lexically and grammatically opposed suffixes. Akhvakh, Lak, Literary Dargwa and some other languages preserve the opposition between Adverbial comparatives (with several oblique stem markers) and Particle comparatives (with a single comparative particle).

The situation in Tsugni Dargwa, which we discussed in Section 4, is generally similar to that in Akhvakh, Literary Dargwa, or Lak. The difference is that in Tsugni, where the initial spatial markers attached to the direct stem in the singular, the opposition of Adverbial and Particle comparatives was only relevant for plural nouns. As soon as standards of comparison in the plural shifted to using the absolutive stem by analogy with the singular forms, the subtle difference between comparative suffixes and comparative particles was effaced, with the result that we no longer have any reason to identify Adverbial comparatives in Tsugni."

6 Results and conclusions

Our pilot research has shown that although most East Caucasian languages have Adverbial Comparative constructions, Particle Comparatives are not uncommon. Based on the presence and origin of Adverbial and Particle comparatives, the languages fall into several groups as shown in Table 3.

In the first group of languages, only Adverbial comparatives are found (Section 5.2). They can be further subdivided as to the basic meaning of the comparative marker (spatial vs. dedicated), but

the modern dedicated markers themselves often go back to former spatial morphemes¹⁶. The main strategy of ND-comparison in these languages is nominalization.

In a few languages, the Particle comparative is the only comparative construction used (Section 5.3). In Tsugni Dargwa and possibly Tabasaran, the comparative particle originates from a spatial marker. In Kajtag, as we have suggested, the comparative particle also goes back to a spatial suffix, but one borrowed from Turkic¹⁷.

There are also two groups of languages where the two types of comparative constructions coexist: one group where the Particle construction developed from the Adverbial construction and another group where it has an alternative origin. In Akhvakh, Khwarshi, Lak and other languages, the Particle comparatives developed from the Adverbial comparatives via broadening of the distribution of the standard marker and syntactic restructuring of the construction (Section 5.5). In most cases, the new comparative particle originates from the combination of an oblique suffix with the relevant locative marker that was already used for comparison.

In Agul, Godoberi, Udi and Archi (Section 5.4), the Particle comparative arose independently. We know of several sources for comparative particles, including the postposition meaning ‘except’ in Agul and the terminative marker in Udi. In Karata, the comparative particle *χoror* is identical to a conditional marker (Magomedova & Khalidova 2001: 281).

The presence of a Particle Comparative correlates with the strategy of ND-comparison. In Section 5.5, we showed that the development of Particle constructions begins in ND-comparatives: languages that use nominalization can do without comparative particles; languages that compare adverbs and other non-declinable words by expanding the distribution of locative suffixes tend to develop Particle comparatives.

Table 3. Types of comparative constructions and distribution of languages

Standard marker		ND-comparison	Languages
Adverbial comparative	Particle comparative		
Group 1. Adverbial comparatives only:			
LOCATIVE		nominalization of the standard of comparison	Avar, Hinuq, Lezgian, Sanzhi Dargwa, Tanti Dargwa, Tsakhur, Tsez (Andi, Batsbi, Rutul, Tindi, Kryz)
DEDICATED			Hunzib, Ingush, Khinalug (Chechen)
Group 2. Particle comparatives only:			
	→ LOCATIVE	(expansion of the standard marker)	Tsugni, Tabasaran
	OTHER		Kajtag
Group 3. Adverbial and Particle comparatives:			
LOCATIVE	→ LOCATIVE	(expansion of the standard marker)	Akhvakh, Bagvalal, Khwarshi, Lak, Literary Dargwa (Chamalal, Bezhta, Budukh)
LOCATIVE DEDICATED	OTHER		Agul, Godoberi, Karata, Udi (Botlikh) Archi

The distribution of different types of comparative construction does not show any obvious connection with genetic grouping. All the cells in Table 3 – whenever they contain more than one language – bring together languages from different subgroups of the East Caucasian family.

¹⁶ This idea has been expressed in Naccarato (2020).

¹⁷ A Turkic suffix is the alternative source of the comparative particle in Tabasaran.

Contact phenomena may play a role in some cases (as we suggested with regard to the influence of Lak on Tsugni), but we cannot establish a clear correspondence between types of comparative structures and geographical zones. Most likely, Particle comparatives have emerged independently in different languages and dialects.

We suggest that there are at least two factors that support this change. The first of these has been discussed in this paper, namely the existence of what we call ND-comparison: speakers need to compare constituents headed by non-declinable words. In a language with only Adverbial comparatives, this is impossible in the absence of additional techniques (such as nominalization).

The second functional advantage of Particle comparatives is the case when a comparative construction has more than one potential comparee – for example, sentences with transitive verbs like (85).

- (85) a. *John loves you more than Mary [loves you].*
b. *John loves you more than [John loves] Mary.*
(Dixon 2005: 19)

Adverbial comparatives can be ambiguous in this case, cf. sentence (65) from Khwarshi. By contrast, the East Caucasian Particle comparatives are clear in their meaning, because the standard of comparison takes the same case as the comparee.

The development from a locative case suffix to a comparative clitic is a case of degrammaticalization (Norde 2009; 2011): a former inflectional element becomes a syntactically independent unit. In all languages where we observed this change, the former case markers remain phonologically bound. In terms of Norde (2009: 152), this case of degrammaticalization can be characterized as deinflectionalization. According to Norde (2009: 152), this is a composite change “whereby an inflectional affix in a specific linguistic context gains a new function, while shifting to a less bound morpheme type”. In those languages where a Particle comparative developed out of a previous Adverbial construction, an inflectional marker (to be more precise, a sequence of two or three suffixes) expanded its distribution and transformed into a clitic particle.

Generally, degrammaticalization is believed to be a rare phenomenon: hence the existence of a set of parallel degrammaticalization processes in several lects of a language family is an interesting case that needs more detailed research.

Transcriptions, glosses and other abbreviations

General

In examples from Tsugni and other Dargwa lects, we use the transcriptions that are generally accepted in East Caucasian studies.

When citing examples from other authors, we have retained their transcriptions with only one exception: we use the symbol ‘:’ instead of the macron to transcribe long vowels and geminated consonants (č:, š:, etc.). We have not changed morpheme boundaries or the interpretation of any morphemes, but have unified synonymous glosses, for example by replacing CONV and CNV with CVB for converb, PRES with PRS for present, etc. If authors separated the nominal root from the suffix with a full stop, we have replaced this with a hyphen. In examples from East Caucasian languages, we have also replaced the term and corresponding gloss ablative (ABL) with relative (EL).

Glosses

1, 2, 3 – 1st, 2nd, 3rd person; I-V – genders; ABS – absolutive; ACC – accusative; ADD – additive particle; ADJ – adjective; ADV – adverb, adverbializer; ALL – allative (orientation); AN – animate; ANTE – localization ‘in front of’; AOR – aorist; APUD – localization ‘at, close to’; ASR – assertive; ATTR – attributive; CARD – cardinal (numeral); CL – nominal class prefix (Landuma); CMP – comparative (degree of adjectives/adverbs);

1 COMP – comparative (standard marker); COND – conditional; CONTR – contrastive; COP – copula; CVant –
2 anterior converb; CVB – converb; D – gender agreement marker (Ingush); DAT – dative; DEF – definite;
3 DEM – demonstrative; DIR – directive (orientation); DIST – distal; DOWN – downward (direction); EL –
4 elative (orientation); EMPH – emphatic; EQUAT – equative; ERG – ergative; ESS – essive (orientation); EVID
5 – evidential; EXST – existential copula; F – feminine; FIN – finalis; FOC – contrastive focus; FUT – future; GEN
6 – genitive; GNT – general tense; HAB – habitual; HITHER – ‘to the speaker’ (direction); HPL – human plural
7 gender; IN – localization ‘inside (a container)’; INCL – inclusive; INDF – indefinite; INF – infinitive; INFR –
8 inferential (tense/evidential); INTER – localization ‘in (a substance); between’; IPFV – imperfective; LAT –
9 lative (orientation); LL – lower level; LOC – general (default) localization; LV – light verb; M – masculine;
10 MSD – masdar (deverbal noun); N – non-human gender; NEG – negation; NMLZ – nominalizer; NOM –
11 nominative; NPL – non-human plural gender; NREF – non-referential; O – object; OBL – oblique stem; OBLQ
12 – oblique (syncretic general non-nominative case of adjectives and participles); PFV – perfective; PL –
13 plural; POS – positive; POSS – possessive preposition; POST – localization ‘behind’; PRET – preterite; PROX
14 – proximal; PRS – present; PST – past; PST.W – past witnessed; PTCP – participle; SBST – substantivizer; SG –
15 singular; SL – same level; SUB – localization ‘under’; SUPER – localization ‘on’; TH – thematic stem; THITHER
16 – direction ‘from the speaker’; TRANS – transitive; UP – upward (direction)

17 Other abbreviations

18 ABS-comparison – comparative construction with the comparee expressed by an absolutive NP; ND-
19 comparison – comparative construction with the comparee expressed by a constituent headed by a
20 non-declinable word (adverb, infinitive, converb); GM – position of a gender marker; OBL-
21 comparison – comparative construction with the comparee expressed by an NP in an oblique case;
22 PN – personal name

23 Appendix: Data sources

Agul	Magometov 1970: 84; Maisak 2008; Maisak 2014: 423; Timur Maisak (p.c.), Solmaz Merdanova (p.c.)
Akhvakh	Creissels 2009
Andi	Salimov 2010: 125, 228
Archi	Chumakina 2024; Marina Chumakina (p.c.)
Avar	Bokarëv 1949b: 165; Forker 2020: 266; Yakov Testeleets (p.c.)
Budukh	Meilanova 1984; Talibov 2006
Bagvalal	Sosenskaja 2001a
Batsbi	Desheriev 1953: 64
Bezhta	Comrie, Khalilov & Khalilova 2015: 269, 292; Khalilova 2016; Zaira Khalilova (p.c.)
Botlikh	Saidova, Abusov 2012
Chamalal	Bokarëv 1949a: 108
Chechen	Nichols 1994; Aliroev 1999: 50
Godoberi	Sosenskaja & Tatevosov 1996
Hinuq	Forker 2013: 708–713
Hunzib	van den Berg 1995: 50
Ingush	Nichols 2011: 161–164, 425–427, 516–517
Kajtag Dargwa	Temirbulatova 2002: 263, 267; Pavel Dolgoplov, Ivan Kalyakin (p.c.)

Karata	Pasquereau 2010: 38, 42, 60; Magomedova & Khalidova 2021; Rashidat Khalidova (p.c)
Khinalug	Desheriev 1959: 25–26; Khvtisiashvili 2013: 95–96; Monika Rind-Pawlowski (p.c.)
Khwarshi	Khalilova 2009: 285–291
Kryz	Authier 2009: 179
Lak	Kazenin 2013: 287–291; Friedman 2020
Lezgian	Haspelmath 1993: 432–434
Literary Dargwa	Abdullaev 1961: 186–187; Abdullaev et al. 2014: 213, 357; Yusupov 2017
Rutul	Ibragimov 2004: 60; Michael Daniel (p.c.)
Sanzhi Dargwa	Forker 2020: 569–570
Tabasaran	Magometov 1965: 335; Babaliyeva 2013
Tanti	Sumbatova, Lander 2014: 73; Lander, Mamaev 2021; Yury Lander, Magomed Mamaev (p.c.)
Tindi	Magomedova 2003
Tsakhur	Sosenskaja 1999
Tsez	Polinsky 2015: 82–84
Udi	Maisak 2018; Timur Maisak (p.c.)

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